

AI-Assisted Mathematical Reasoning and Statistical Outlier Detection

Independent Mathematics & AI Portfolio Project — 2026

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Project status: Independent project created in 2026. It is not represented as a university project, published paper, or peer-reviewed research.

Abstract

This project demonstrates an interdisciplinary approach combining mathematical reasoning, statistics, computer science, and contemporary AI evaluation practices. A reproducible synthetic dataset is constructed with a quantitative relationship between study hours and assessment scores, followed by the deliberate injection of unusual observations. Classical IQR-based outlier detection is compared with Isolation Forest. The project then connects the same verification principles to the evaluation of AI-generated mathematical answers, emphasizing independent checking, error identification, assumptions, and structured scoring.

1. Introduction

Reliable mathematical reasoning requires more than obtaining a final number. A strong solution should use valid formulas, consistent assumptions, correct intermediate calculations, and a conclusion that follows from the evidence. These principles are also important when evaluating AI-generated quantitative responses. This project uses outlier detection as a concrete example of how mathematical and statistical reasoning can be translated into a reproducible computational workflow.

2. Objectives

- Apply descriptive and inferentially useful statistical concepts to a controlled dataset.
- Demonstrate the IQR method for interpretable univariate outlier screening.
- Demonstrate Isolation Forest for multivariate anomaly detection.
- Compare detection outputs using precision, recall, F1-score, and accuracy.
- Develop a practical framework for verifying AI-generated mathematical reasoning.

3. Mathematical and Statistical Methodology

Interquartile Range (IQR): $IQR = Q3 - Q1$. The standard screening bounds used here are $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$. Values outside these limits are flagged.

Z-score: $z = (x - \mu) / \sigma$. Z-scores are useful for expressing how far an observation lies from the sample mean in standard-deviation units.

Correlation: Pearson correlation is used as a descriptive measure of linear association between study hours and assessment score.

Isolation Forest: observations that can be isolated with relatively short partitioning paths are treated as more anomalous. The method is designed for unsupervised outlier detection.

4. Dataset Design

The dataset contains 130 observations: 120 baseline observations and 10 deliberately injected anomalies. The variables are `study_hours` and `assessment_score`. The data are synthetic and reproducible using random seed 42.

Observed Pearson correlation in the generated dataset: **-0.316**.

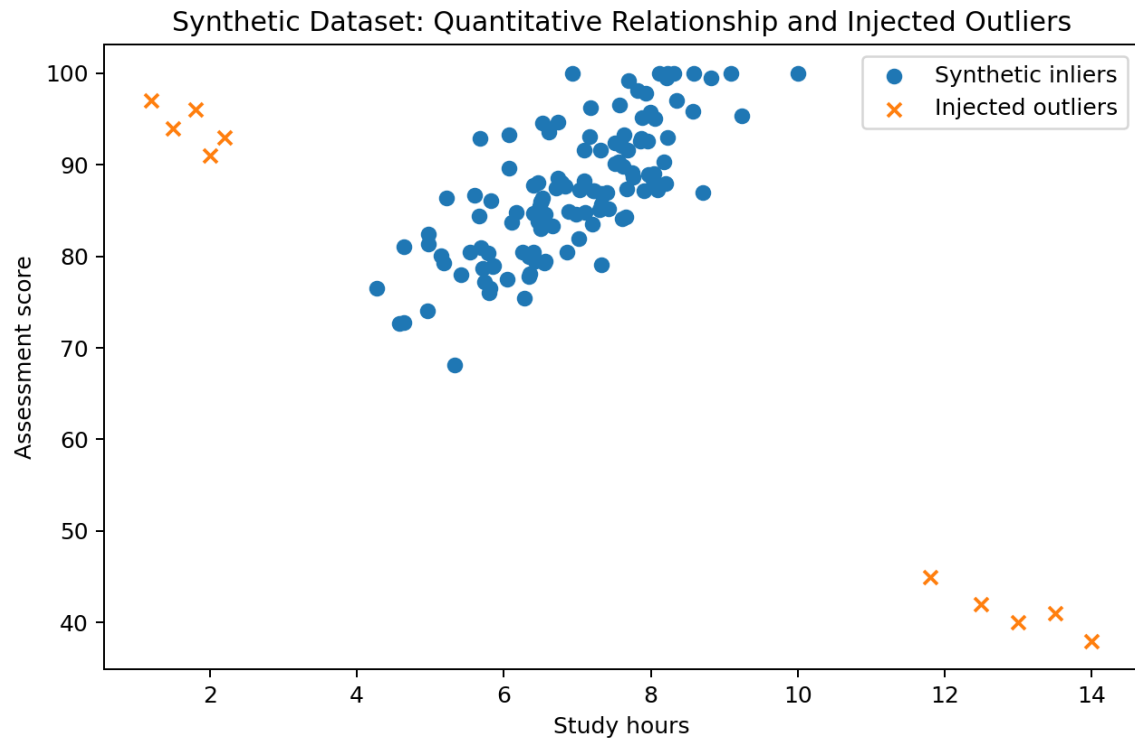


Figure 1. Synthetic quantitative relationship with intentionally injected anomalies.

5. Outlier Detection Results

Method	TP	TN	FP	FN	Precision	Recall	F1	Accuracy
IQR	10	120	0	0	1.00	1.00	1.00	1.00
Isolation Forest	8	118	2	2	0.80	0.80	0.80	0.97

The results illustrate an important methodological point: no single outlier detector should automatically be treated as ground truth. The IQR method is transparent and easy to explain, but it examines each variable separately. Isolation Forest considers the joint feature space and can therefore identify observations that are unusual because of their combination of values. The choice of method should reflect the data-generating context.

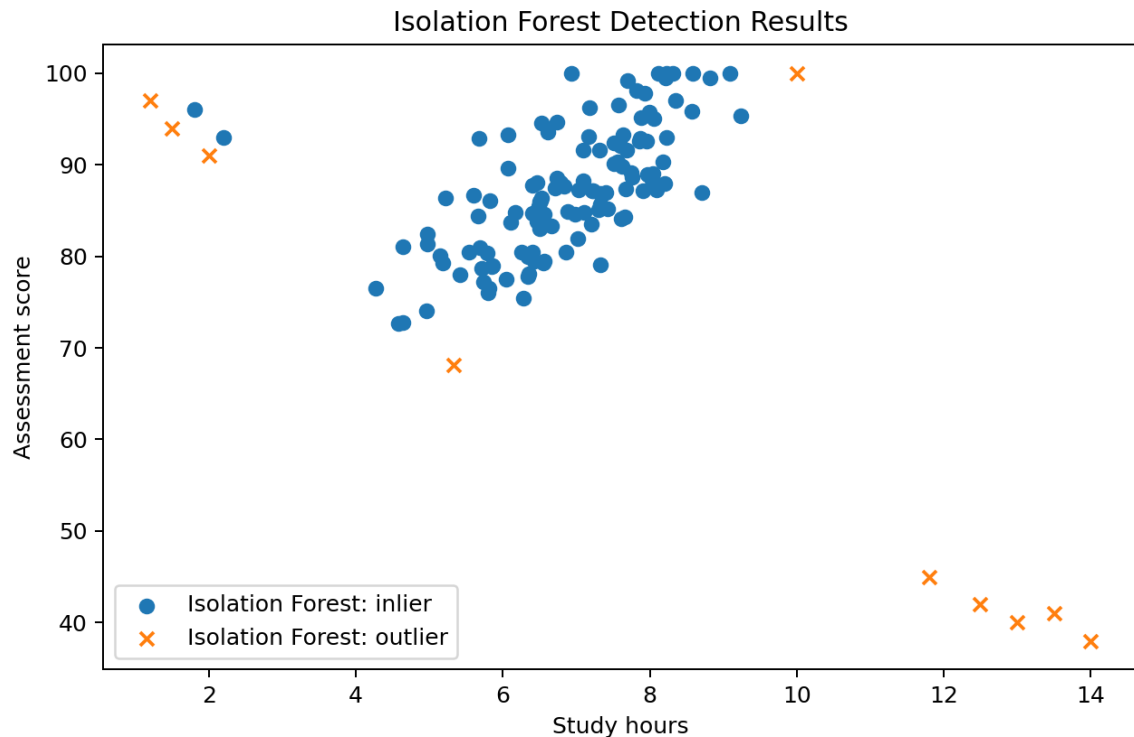


Figure 2. Isolation Forest classification of the synthetic dataset.

6. AI-Assisted Mathematical Evaluation Framework

- **Prompt design:** Write a precise problem with sufficient constraints and an unambiguous expected output.
- **Independent solution:** Solve the problem without relying on the model's proposed answer.
- **Step verification:** Check formulas, arithmetic, transformations, units, and logical implications.
- **Error classification:** Separate calculation errors, conceptual errors, missing assumptions, and communication issues.
- **Ranking:** Score responses on correctness, completeness, rigor, relevance, and clarity.
- **Feedback:** Explain the exact point of failure and provide the mathematically justified correction.

7. Example Evaluation Rubric

Criterion	Weight	Evaluation question
Correctness	40%	Is the final answer mathematically correct?
Reasoning	25%	Are the intermediate steps valid and logically connected?

Criterion	Weight	Evaluation question
Completeness	15%	Are necessary assumptions and steps included?
Clarity	10%	Can another reader follow the reasoning?
Efficiency	10%	Is the approach appropriately concise without losing rigor?

8. Discussion

The project demonstrates how an academic foundation in mathematics and statistics can be extended into modern AI evaluation work. Classical mathematical rules provide transparent checks, while computational methods provide scalable pattern detection. In an AI-training context, the human evaluator remains responsible for validating whether a model's reasoning is actually correct rather than merely plausible.

9. Limitations

- The dataset is synthetic and intentionally designed to contain anomalies.
- Injected anomalies do not represent real-world measurement error or naturally occurring rare events.
- Outlier detection is sensitive to method choice, parameters, feature scaling, and data distribution.
- An outlier is not necessarily an incorrect observation; it may be a legitimate rare case.

10. Conclusion

Mathematical expertise is valuable in AI evaluation because reliable model assessment requires independent reasoning, precise calculations, and the ability to distinguish a correct argument from a plausible-looking one. By combining IQR-based statistical screening with Isolation Forest and a structured AI-response evaluation rubric, this project demonstrates a practical bridge between mathematical sciences and contemporary AI quality work.

11. References

1. Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. 2008 Eighth IEEE International Conference on Data Mining, 413–422.
2. Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2012). Isolation-Based Anomaly Detection. ACM Transactions on Knowledge Discovery from Data, 6(1).
3. scikit-learn documentation: Outlier Detection and Isolation Forest. The documentation describes Isolation Forest as an unsupervised outlier-detection method based on isolation/path length.
4. Standard statistical definitions and methods for quartiles, IQR, z-scores, correlation, precision, recall, and F1-score.

Author Note

This is an independently created portfolio project prepared to demonstrate mathematical, statistical, computational, and AI-evaluation skills. It should be presented to employers as an independent project rather than as a university assignment or peer-reviewed publication.